LESSON 1.3

Math is In My World

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can represent a real-world situation with a tape diagram or table and use decimal operations to solve it.

WHY THIS MATTERS Math lives in your world — in coins on a table and tram cars over a valley. Today you visualize real situations, pick tools like tape diagrams and tables, and let decimal arithmetic do the heavy lifting. The tram must carry 4,000 passengers over the valley. Each ride carries 80 passengers and takes 12.5 minutes. How many rides are needed, and how long will all of the rides take?

REMEMBER

Math is In My World

Math Practices: Visual Models & Tools The book opens with a warm-up — about how many coins are in the picture? — then rides an aerial tramway that must move 4,000 passengers.

1. Select the mathematical tool (tape diagram, number line, grid) that best represents the quantities.
2. Label all parts and units clearly.
3. Connect visual relationships directly to numerical expressions

Words: representation *(representación)* · tape diagram *(diagrama de cintas)* · tool *(herramienta)* · ordered pair *(par ordenado)* · round trip *(viaje de ida y vuelta)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each expression by whether it equals 125.

Groups: Equals 125 · Does NOT equal 125

_____ 12.5 × 10

_____ 1.25 × 100

_____ 62.5 × 2

_____ 12.5 × 100

_____ 1.25 × 10

_____ 12.5 + 10

HINT Multiplying by 10 moves the decimal point one place; by 100, two places.

2 **Complete the table. Show how you found each value.**

Each tram ride takes 12.5 minutes. Complete the table of total minutes.

Rides	Expression	Total minutes
10		
20		
50		

HINT Use the first row to build the others: 20 rides is double 10 rides.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

What is 1/2 of 60?

(A) 20
(B) 30
(C) 40
(D) 120

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

4 **Circle the letter of the best answer. Show how you know.**

What is 1/4 of 20?

(A) 4
(B) 5
(C) 16
(D) 80

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

5 **Circle the letter of the best answer. Show how you know.**

You multiply a whole number by a fraction smaller than 1. The answer is:

(A) Smaller than the number you started with
(B) Bigger than the number you started with
(C) Exactly the same
(D) Always zero

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Find a card where you and your partner chose different operations. What is it about the situation — not the numbers — that tells you whether division or multiplication helps?

SENTENCE STARTER This situation needs ___ because we are ___ a total, not ___ it.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which statement best shows what this lesson means by "math is in my world"?

(A) We visualize real problems, use tools to show relationships among quantities, and choose our tools strategically

(B) Real-world problems can only be solved with a calculator

(C) Tables and diagrams are decorations we add after solving

(D) Every problem must be solved with the same one tool

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

It takes 35 minutes for one tram car to make the 5-mile round trip, from valley to mountain top and back down — including time for passengers to disembark and embark at each station. You want to know how many round trips a single tram car completes in 10 hours. The tram problem asks how many round trips fit in 10 hours. What is the FIRST thing you must find before you can read the answer off your table?

(A) How many miles the tram travels in total

(B) How many passengers ride each trip

(C) How many minutes are in 10 hours, so the units match the table

(D) The price of a ticket

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

It takes 35 minutes for one tram car to make the 5-mile round trip, from valley to mountain top and back down — including time for passengers to disembark and embark at each station. You want to know how many round trips a single tram car completes in 10 hours. Using the table (35→1, 175→5, 350→10, 525→15, 700→20 minutes→trips), why does the answer end up being 17 round trips, not 18?

(A) 18 trips is closer to 600 minutes, so it is the better estimate

(B) 17 trips take 595 minutes, which fits within 600, but an 18th trip would need 630 minutes, past the limit

(C) A round trip can be counted as complete even if it is only partly finished

(D) The table only lists rows up to 20, so 17 is the safe middle choice

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

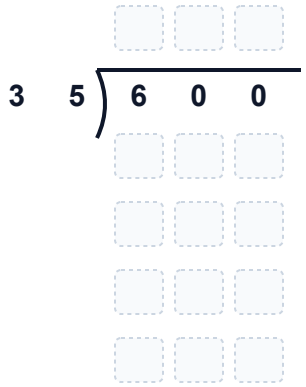
It takes 35 minutes for one tram car to make the 5-mile round trip, from valley to mountain top and back down — including time for passengers to disembark and embark at each station. You want to know how many round trips a single tram car completes in 10 hours. 600 ÷ 35 ≈ 17.14. Why is it risky to round this quotient down to 17 WITHOUT also checking the table?

(A) Rounding down happens to work here, but only the table confirms 17 × 35 stays under 600 while 18 × 35 does not

(B) Rounding down never works for time problems

(C) 17.14 should round up to 18 because .14 rounds upward

(D) The table and division always disagree with each other



EXPLAIN YOUR THINKING

Find a card where you and your partner chose different operations. What is it about the situation — not the numbers — that tells you whether division or multiplication helps?

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